Normalization and Evaluation of Proteomics Data with PRONE: Methods, Metrics and Selection Guide

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The best normalization method is dataset-specific: no single method wins everywhere, and low intragroup variation alone does not identify it. This guide summarizes the PRONE benchmark (17 normalization methods, 2 batch-correction methods, 6 spike-in and 3 biological datasets) and how to choose among methods.

Scope and sources

This guide is based on Arend et al. (2025) and its Supplementary Methods, plus the PRONE documentation (Bioconductor package, version 1.0.4) and function reference pages.

- Method descriptions come from the Supplementary Methods. Where the supplement is silent, a statement is marked as general knowledge or as the author's inference.
- PCV and PEV are not defined in the paper, the supplement or the PRONE reference pages. Only PMAD is defined (main text, Methods).
- Supplementary figures were read through their captions only; the figure images were not inspected.
- Datasets: six spike-in datasets (dS1–dS6; UPS1, E. coli or yeast spike-ins; four label-free, two TMT) and three biological datasets (dB1 and dB2 TMT, dB3 label-free).

Normalization methods

RobNorm, LoessF, the Normics methods and the reference-based regression methods (Rlr, RlrMA) performed best in the spike-in benchmark; EigenMS and MAD performed worst downstream. Methods are grouped as in the paper's Table 2.

Method	What it does	Why use it	Advantages	Disadvantages and evidence
Mean / Median	Shifts each sample to a common mean or median	Corrects loading differences	Simple, transparent	Assumes a stable background. Asymmetric spike-in shifts contaminate the factors; higher FPR in dS5 and dS6
GlobalMean / GlobalMedian	Scales by the mean or median of per-sample total intensities	Corrects total protein amount and labeling efficiency	Simple, clear rationale	Same assumption as above; the two variants gave near-identical results
MAD	Median-centers and scales so median and spread are constant across samples (NormalyzerDE function)	Corrects scale and dispersion differences	Robust scale estimate	Compresses variance and logFCs. Lowest PMAD (0.016–0.088 vs 0.052–0.341 for log2) but poor F1 and almost no DE proteins at logFC >1 in biological data
NormicsMedian	Ranks proteins by likelihood of being non-DE (expression-adjusted variance and mean correlation with all other proteins), then median-normalizes using only those; parameters are applied to all proteins	Estimates factors from a stable subset	Not distorted by DE or variable proteins; top group in F1	New, little independent validation; lower Jaccard overlap in dB1
Quantile	Ranks values per sample, averages by rank, replaces values with the rank average	Forces identical distributions	Strong alignment	Assumes most proteins are unchanged; higher FPR in dS6
Rlr	Robust linear regression of each sample on a reference (median of all samples)	Corrects bias linear in intensity	Single reference	Assumes linear bias; F1 above 0.6 in dS2

Method	What it does	Why use it	Advantages	Disadvantages and evidence
RlrMA	Rlr on MA-transformed data (A = median sample, M = difference)	Same	Top group in F1	As above
RlrMACyc	MA transformation and pairwise normalization of all sample pairs, 3 cycles	Same, iterative	Converges in 3 cycles	One missing value in a protein-sample makes the whole protein NA after normalization; F1 below 0.2 in dS2
LoessF / LoessCyc	Local (non-linear) regression; F is the loess analogue of RlrMA, Cyc of RlrMACyc; both via limma normalizeCyclicLoess	Corrects intensity-dependent, non-linear bias	LoessF among the best (F1 not significantly different from RobNorm, p = 0.1)	LoessCyc propagates missing values like RlrMACyc and was weak in dS2
TMM	edgeR scaling factors with double trimming on M and A values	Compositional bias	Excludes extreme proteins from the factors	Lower F1 in dS1–dS2; behaves like plain log2 in biological data (over 125 DE proteins in dB1) and does not reduce PMAD
VSN	Parametric transformation fitted by maximum likelihood so variance is independent of mean intensity	Heteroscedasticity, outliers	Principled, widely used	Sensitive to lts.quantile: under 25 DE proteins in dB1 with the default versus a normal count with 0.5
NormicsVSN	VSN fitted only on the selected non-DE proteins, then applied to all	Combines both ideas	Top-tier F1	Same caveats as NormicsMedian plus the quantile parameter

Method	What it does	Why use it	Advantages	Disadvantages and evidence
EigenMS	ANOVA on the experimental factors, then SVD on the unexplained variation to remove systematic trends	Structured, batch-like bias	Preserves group-level differences	Needs more than one valid value per protein per group, otherwise the protein becomes all NA. Uses the design, so overfitting is plausible (author's inference). Highest FPR, lowest F1 (median below 0.4), inflated DE counts
RobNorm	Density-power-weight model that estimates sample effects and down-weights outliers	Heterogeneous samples such as different tissues	Does not assume similar intensities across samples; best median F1; background logFC closest to 0	Docs advise gamma.0 of 0.1 or 0.5 for fewer than 40 samples

Batch-effect correction (TMT)

In the two TMT biological datasets, limBE removed batch effects more effectively than IRS while keeping the biological signal. PRONE applies batch correction either before or after normalization, at the user's choice.

- **IRS (internal reference scaling).** Computes each protein's geometric mean across the reference samples, derives one scaling factor per TMT plex (geometric mean divided by that plex's reference intensity), and multiplies the plex by it. It needs a reference channel in every plex and takes raw input.
- **limBE (limma removeBatchEffects).** Fits a linear model with batch and treatment terms and removes only the batch component. Unlike a model that estimates both effects at once, it cleans the data itself. Risk (general knowledge): if batch and condition are confounded, the biological signal can be removed too.

Evaluation metrics

Intragroup variation metrics alone are misleading: the methods with the lowest PMAD (MAD) and highest correlation (EigenMS) perform poorly downstream. Combine visual checks, intragroup metrics, batch assessment and DE-based metrics.

Metric or plot	What it assesses	Better values	Strengths	Limitations
Boxplots / densities	Alignment of sample intensity distributions	Distributions aligned more closely after normalization	Fast QC; shows outlier samples	Qualitative; alignment does not prove preserved biology
PCA / hierarchical clustering	Dominant structure and batch	Samples cluster by biological group, not technical factors (use shape_by for batch)	Reveals batch and outliers at a glance	Subjective; depends on the components shown
PMAD	Pooled median absolute deviation of replicates within groups	Lower	Paper's main metric; more robust to outliers than PCV and PEV	Rewards scale compression: MAD wins it yet performs poorly downstream
PCV / PEV	Pooled coefficient of variation / pooled variance within groups	Lower	Standard Normalyzer-style summaries	Not defined in the docs or supplement; more outlier-sensitive than PMAD (main text)

Metric or plot	What it assesses	Better values	Strengths	Limitations
Intragroup Pearson correlation	Replicate similarity	Higher	Scale-free	Weakly discriminating: most methods are identical to three decimals (Supp. Table 3), as expected because correlation is invariant to per-sample linear transformations. EigenMS has the highest values yet the worst F1
Silhouette coefficient (PC1–3)	How well samples fit their batch or condition group	Batch: toward 0 or below (batches mixed). Condition: toward 1	Quantitative batch assessment	Needs batch and condition not confounded; only 3 PCs
AUC	Ranking of spike-in against background proteins	Higher	Familiar	Uninformative here: all methods exceeded 0.75 despite large FPR differences (class imbalance)
FPR / TPR	Error rates on spike-in data	Low FPR, high TPR	Exposes false positives AUC hides	Depends on the DE method and thresholds
F1 score	Harmonic mean of precision and recall	Higher (range 0 to 1)	Handles class imbalance; penalizes false positives	Needs ground truth; varies by dataset (RobNorm: 0.51 in dS1, 0.93 in dS6)

Metric or plot	What it assesses	Better values	Strengths	Limitations
logFC accuracy	Observed minus theoretical logFC for spike-ins; background logFC	Spike-in close to theoretical; background centered at 0	Detects variance and effect compression	Most methods underestimate spike-in logFCs
Number of DE proteins	Size of the DE result	Not more is better	Needs no ground truth	Large counts may be false positives (TMM, EigenMS, log2 in dB1)
Jaccard, UpSet, consensus set	Agreement of DE sets across methods	Higher agreement means more robust	Shows how much the normalization choice matters	Agreement is not correctness
KEGG enrichment overlap	Functional consistency of DE results	Higher	Checks biological conclusions	Depends on the DE list and database

How to choose a normalization

Choose on criteria fixed beforehand, using intragroup metrics only to screen and DE-based evidence to decide.

1. **Preprocess.** The paper keeps proteins with at least 80% valid values, removes contaminants and reverse hits, and runs POMA outlier detection before normalizing.
2. **Run many methods plus a log2 baseline.** For TMT data add batch correction: prefer limBE; use IRS only if every plex has reference channels.
3. **Screen visually and with silhouette.** Check distribution alignment, whether PCA groups samples by condition, and whether batch silhouette falls toward 0 or below.
4. **Use PMAD as a sanity check, not a ranking.** A method that wins by shrinking the scale (MAD) is a warning sign.
5. **Compare downstream DE** across candidates with one fixed DE method: DE counts, volcano plots, Jaccard and UpSet overlap, enrichment overlap, and the behavior of known positive-control proteins.
6. **Use F1 and FPR, not AUC, if any ground truth exists** (spike-ins or validated markers).

7. **Do not pick the method that gives the most DE proteins.** Selecting on the outcome is circular.

Matching method to data:

- Clean, low-noise data: most methods agree, so a consensus DE set is reasonable (the authors' hypothesis, not yet validated).
- Heterogeneous, noisy data: the choice matters a lot and needs validation, for example by enrichment analysis.
- Many missing values: prefer non-cyclic methods (LoessF, RlrMA, RobNorm, Normics) over RlrMACyc and LoessCyc.
- Many proteins expected to change globally: avoid methods built on the assumption that most proteins are unchanged (Mean, Median, Global*, Quantile).

Limitations of the evidence

- **Spike-in design.** Spike-in data lack biological variation, and spike-in proteins are all shifted in one direction, which violates the symmetry assumption of the simple shift methods. dS4 has a 30.9% spike-in rate (1530 of 4951 proteins) and dS5 21.67%, so a large share of proteins truly change; poor results of Mean, Median and Global* there may reflect the design (author's inference).
- **Small designs.** dS4 has two conditions with three replicates each. dS2 was restricted to higher UPS1 concentrations because of missing values.
- **DE methods and thresholds.** Only limma and ROTS were compared. F1 depends on thresholds (adjusted p below 0.05; absolute logFC above 1 in biological data), and the ranking shifted somewhat with ROTS.
- **Biological datasets.** Three datasets without ground truth, so conclusions there rest on inference. The suspicion that EigenMS and TMM DE counts are false positives is a hypothesis in the paper.
- **Supplement inconsistencies.** The supplement states VSN was run with normalizeCyclicLoess, which looks like a copy error (author's reading). Supp. Table 4 lists 4 POMA outliers for dB2 while its caption says none were excluded. The paper used lts.quantile 0.9 for Normics while the PRONE reference shows 0.8 as default.

Example R workflow

Function names are from the PRONE reference index; arguments are abbreviated, so check the documentation for the full signatures.

```
library(PRONE)

# Normalize with several methods (stored as assays); batch correction can be
chained with combination_pattern = "_on_"
se <- normalize_se(se, c("RobNorm", "LoessF", "NormicsVSN", "limBE"))

# Visual checks
plot_boxplots(se); plot_densities(se); plot_PCA(se, color_by = "Condition")

# Intragroup variation (diff = TRUE shows reduction relative to log2)
plot_intragroup_PMAD(se, diff = TRUE)
plot_intragroup_correlation(se)

# Downstream DE and robustness
se <- run_DE(se, DE_method = "limma")
plot_overview_DE_bar(se); plot_upset_DE(se); plot_jaccard_heatmap(se)

# Spike-in data only (ground truth)
stats <- get_spiked_stats_DE(se)
plot_stats_spiked_heatmap(se); plot_ROC_AUC_spiked(se)
```

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